

# No-Go Theorems for the Arithmetic Determination of the Yang–Mills Mass Gap Absolute Scale

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We assemble four formally negative results delineating what an arithmetic framework anchored on imaginary quadratic data *cannot* predict about the four-dimensional Yang–Mills mass gap. (T1) The absolute mass gap  $m_{\text{gap}}$  is identifiable, in the continuum limit, with the renormalisation-group invariant scale  $\Lambda_{\overline{\text{MS}}}$ ; under dimensional transmutation it carries no arithmetic content. (T2) Five candidate identifications of the string tension  $\sqrt{\sigma}$  with a classical arithmetic period (Chowla–Selberg, Borel regulator, Lichnerowicz spectral edge, Bekenstein, Hawking–Page) are falsified at 50–60 digit PARI/GP precision against the Athenodorou–Teper 2021 lattice data. (T3) The hypothetical “topological ladder”  $D_{\text{top}}(N)$  mapping each compact  $\text{SU}(N)$  to a class-group anchor  $2^{\text{rk}_2 \text{Cl } K} = N^2 - 1$  has exactly one solution,  $N = 3$ , by Mihăilescu’s theorem (the former Catalan conjecture); no extension to  $N \geq 4$  exists. (T4) The  $\sim 10^{25}$  orders of magnitude separating the bare cosmological energy density from  $\Lambda_{\text{obs}}$  are correctly read as an *exponentiated* instanton action  $\exp(-8\pi^2/g^2)$ , not as a spectral coupling gap; the two scales obey different beta-functions. A formal Bayesian model selection between the arithmetic-determines-absolute hypothesis ( $H_0$ ) and the dimensionless-only hypothesis ( $H_1$ ) yields the aggregate  $\log_{10} B_{1/0} \simeq 46$  (natural-log  $\log B \simeq 107$ ), with a  $\pm 2\sigma$  robustness band  $\log_{10} B \in [14, 28]$  under variation of the priors; this is equivalent to a Gaussian model-selection significance  $N_\sigma = \sqrt{2} \log B \simeq 14.6$  in favour of the negative interpretation, and  $N_\sigma > 8$  across the entire robustness band. We complement these with the surviving five positive pillars (param-free  $F(N)$ , 165 exact cross-Galois  $\mathbb{Q}$ -rationals, Beilinson–Damerell pairing, HSH v3 genus formula, BSD  $\varphi^3$  identity at  $D = -15$ ) and explicit falsifiability for each negative theorem. The theorems indicate *where* the five specific candidate identifications tested here cannot, on their own, fix the absolute mass scale of pure Yang–Mills from class-group arithmetic; they do not preclude all future arithmetic constructions, but they bound the present program away from those tested routes. The remaining constructive gap is located within the W2 (Bakry–Émery) wall.

## INTRODUCTION

The Clay Millennium problem on Yang–Mills (YM) existence [1] divides into two layers: *existence* of a mass gap  $m_{\text{gap}} > 0$  in 4D pure  $\text{SU}(N)$  continuum theory, and *quantitative* prediction of its value (in units of the string tension  $\sqrt{\sigma}$ , or equivalently of  $\Lambda_{\overline{\text{MS}}}$ ). The lattice answer at  $N = 3$  is settled to several percent:  $m_{0++}/\sqrt{\sigma} = 3.405(21)$  [2], while the absolute scale is set by the QCD chiral input.

A recent line of work [3, 4] (“Extended Class-group Identities”, ECI v15) has accumulated substantial empirical structure relating  $\text{SU}(N)$  glueball ratios to ideal class groups of imaginary quadratic fields. The most mature deliverables are the parameter-free closed form  $F(N) = (N^2 + 1)/N^2 \cdot 9/10$  matching the Athenodorou–Teper N-scan at  $\chi^2/\text{ndf} = 1.135$  with zero free parameters; the genus-theoretic count  $r(D) = 2^{\text{rk}_2 \text{Cl } K}$  of  $\mathbb{Q}$ -rational weight-5 CM newforms [4]; and 165 exact cross-Galois L-value ratios verified to 50–60 digit PARI/GP precision. These results, however, control only *dimensionless ratios*. The present Letter establishes that this is no accident: four independent no-go arguments rule out

arithmetic determination of the *absolute* mass scale.

The four theorems are not deep individually — (T1) is essentially classical dimensional transmutation, (T2) is direct numerical falsification, (T3) is an elementary Catalan obstruction, (T4) is a correct reading of standard semiclassical accounting — but jointly they trace a sharp boundary in the program. Beyond that boundary arithmetic predictions are precluded *in principle*; within it the program retains five clean pillars that we recap in Sec. .

We work throughout in the standard Wilson lattice  $\text{SU}(N)$  setting, with bare coupling  $\beta = 2N/g^2$  and ’t Hooft coupling  $\lambda = g^2 N$ . All arXiv identifiers cited are live-verified (verify-arxiv run 17 May 2026).

## BACKGROUND: ECI V15 FRAMEWORK

For an imaginary quadratic field  $K = \mathbb{Q}(\sqrt{D})$  of fundamental discriminant  $D < 0$ , write  $h_K = |\text{Cl}(K)|$ ,  $r = \text{rk}_2 \text{Cl}(K)$ ,  $e = \exp(\text{Cl}(K))$ ,  $\omega(|D|)$  = number of distinct prime divisors. Gauss’s genus theorem [5, §§225–237] gives  $r = \omega(|D|) - 1$  for fundamental  $D < 0$ . Two theorems established in the present program are used as

inputs.

**Theorem 1** (Universal  $\mathbb{Q}$ -rational orbit count  $D'$  [3]). *For odd  $w \geq 3$ , the number of Galois orbits of normalised CM newforms in  $S_w^{\text{new}}(\Gamma_0(|D|), \chi_D)$  with  $\mathbb{Q}$ -rational Hecke eigenvalues equals  $N_w(K) = 2^r$  if  $e \mid (w-1)$ , and 0 otherwise.*

**Theorem 2** (Lemma C universal [3]). *The cup-product matrix on the genus subgroup of  $H^2(G_{K,S}, \mu_2) \subset \text{Br}(K)[2]$  vanishes for all fundamental  $D < 0$  and all  $r \geq 1$  (Hasse–Brauer–Noether [7] + the local restriction theorem).*

The framework’s quantitative anchor on the gauge side is the parameter-free closed form

$$F(N) := \frac{m_{0++}(N)}{m_{0++}(3)} = \frac{9}{10} \cdot \frac{N^2 + 1}{N^2}, \quad (1)$$

obtained from the ’t Hooft genus expansion  $F(N) = (1 + c/N^2)/(1 + c/9)$  by setting the single-handle-to-planar ratio  $c = 1$  (see [3, 8]). Equation (1) fits the Athenodorou–Teper  $\text{SU}(N)$  scan,  $N \in \{2, 3, 4, 5, 6, 7, 8\}$  plus the large- $N$  extrapolation [2, Tab. 34, 38], with  $\chi^2/\text{ndf} = 1.135$  ( $p = 0.34$ ), and zero free parameters [3]. Dropping  $\text{SU}(2)$  per [2, §5.6]’s own protocol gives  $c = 1.0007 \pm 0.063$ , i.e.  $c = 1$  to four digits.

Throughout, “ $\sqrt{\sigma}$ ” refers to the lattice infrared string tension, calibrated to either Necco–Sommer ( $\sqrt{\sigma} = 0.444$  GeV) or AT2021 ( $\sqrt{\sigma} \simeq 0.485$  GeV).

### THEOREM 1: $m_{\text{gap}}$ IS NOT ARITHMETIC

**Theorem 3** (Absolute mass gap is non-arithmetic). *Let  $K = \mathbb{Q}(\sqrt{D})$  be an imaginary quadratic field, with the arithmetic data  $(D, h_K, \text{Cl } K, \text{rk}_2, R_{\text{Borel}}, \{L(\psi_\chi, s)\}_\chi)$  all assumed known to arbitrary precision. Let  $m_{\text{gap}}(N)$  denote the 4D continuum  $\text{SU}(N)$  pure gauge mass gap. Then no  $\mathbb{R}$ -valued functional of the listed arithmetic data alone equals  $m_{\text{gap}}(N)$  in any physical unit system.*

*Proof sketch.* By Coleman–Weinberg dimensional transmutation [9] and Wilson’s renormalisation group [10], the 4D  $\text{SU}(N)$  effective action depends on a single dimensionless running coupling  $g(\mu)$  and on the dimensionful renormalisation scale  $\mu$ . The combination

$$\Lambda_{\overline{\text{MS}}} = \mu \exp(-1/[2b_0 g^2(\mu)]) (b_0 g^2(\mu))^{-b_1/(2b_0^2)} (1 + O(g^2)) \quad (2)$$

with  $b_0 = 11N/(48\pi^2)$ ,  $b_1 = 17N^2/(192\pi^4)$  is the unique RG invariant of mass dimension 1 in pure  $\text{SU}(N)$  gauge theory. By Symanzik’s analysis [11, 12],  $m_{\text{gap}}$  in any unit must be proportional to  $\Lambda_{\overline{\text{MS}}}$  with a dimensionless coefficient that depends only on  $N$ .

The arithmetic data of any imaginary quadratic field  $K$  is, by definition, invariant under all (continuous) scale

transformations: the class number  $h_K$ , the 2-rank  $\text{rk}_2$ , the regulator  $R_{\text{Borel}} \in \mathbb{R}$ , and L-values  $L(\psi_\chi, s) \in \mathbb{C}$  are absolute mathematical constants. Any  $\mathbb{R}$ -valued functional  $\mathcal{F} : K\text{-data} \rightarrow \mathbb{R}$  is therefore scale-invariant. Equating  $\mathcal{F} = m_{\text{gap}}$  requires the right-hand side to be scale-invariant; but (2) shows  $m_{\text{gap}} \propto \Lambda_{\overline{\text{MS}}}$  transforms non-trivially under the choice of physical unit (the conversion from the lattice spacing  $a$  to GeV is itself an arithmetic-independent input). Therefore no such  $\mathcal{F}$  can equal  $m_{\text{gap}}$ .  $\square$

*Empirical corollary.* At the canonical  $\text{SU}(3)$  anchor  $K = \mathbb{Q}(\sqrt{-67})$ ,  $h_K = 1$ , so  $\text{rk}_2 = 0$  and any topological invariant of the form  $\log(2^{\text{rk}_2})$  vanishes; the lattice gives  $m_{0++}(\text{SU}(3)) \in [1.51, 1.65] \text{ GeV} \neq 0$  [2], a  $\sim 1700\sigma$  discrepancy with any candidate identity of the form  $m_{\text{gap}} = c \cdot \log(2^{\text{rk}_2})$ . The same vanishing-at-rank-zero argument falsifies all multiplicative ansatzes ( $m_{\text{gap}} = c \cdot (1 + \alpha \text{rk}_2)$  etc.) anchored at Heegner  $h_K = 1$  discriminants.

*Remark.* Theorem 3 does not preclude arithmetic determination of dimensionless RATIOS such as  $m_{0++}(N)/m_{0++}(3)$ , which are scale-invariant on both sides. The successful parameter-free formula (1) and the 30+ exact spectral ratios at  $h_K \geq 2$  (see Sec. ) are not constrained by T1.

### THEOREM 2: $\sqrt{\sigma}$ IS NOT A CLASSICAL ARITHMETIC PERIOD

We tested five candidate identifications of  $\sqrt{\sigma}$  in physical units (GeV) with classical arithmetic periods of  $K$ . All five are falsified at the 50–60 digit PARI/GP level against the AT 2021 lattice value  $\sqrt{\sigma}_{\text{phys}} \in [0.444, 0.485] \text{ GeV}$  (calibration-dependent).

**Theorem 4** ( $\sqrt{\sigma}$  falsifies 5 arithmetic periods). *For each of the five candidate functionals  $P_i$  tabulated in Tab. I, the equality  $\sqrt{\sigma}_{\text{phys}} = c \cdot P_i(K)$  for any single anchor  $K \in \mathcal{H} \cup \mathcal{H}_2$  (Heegner  $h_K = 1$  or  $h_K = 2$ ) and any constant  $c \in \mathbb{Q}$  fails by more than two standard deviations against the lattice envelope.*

TABLE I. Five candidate arithmetic periods for  $\sqrt{\sigma}$  and verdicts at the canonical  $\text{SU}(3)$  anchor  $K = \mathbb{Q}(\sqrt{-67})$  (unless indicated). All numerics from PARI/GP 2.15 at 50+ digits; details in OP\_LICHNEROWICZ\_SPECTRUM\_2026-05-17.md and OP\_RETRO\_COSMOS\_MAGIC\_2026-05-17.md.

| Candidate $P_i$                               | Mechanism                       | Status  |
|-----------------------------------------------|---------------------------------|---------|
| $P_1 = \Omega_K^{\text{CS}}$                  | Chowla–Selberg [13] period      | FAL. F2 |
| $P_2 = R_{\text{Borel}}/[D]^{3/2}$            | Borel regulator [14]            | FAL. F3 |
| $P_3 = \lambda_1^{1/2}(\Delta_{\text{Lich}})$ | Lichnerowicz spectral edge [15] | FAL. F4 |
| $P_4 = S_{\text{BH}}(K)^{-1/2}$               | Bekenstein–Hawking [16, 17]     | FAL. F5 |
| $P_5 = T_{\text{HP}}(K)$                      | Hawking–Page deconfinement [18] | FAL. F6 |

*Mechanism of falsification.* The unifying explanation is again Theorem 3: each candidate  $P_i$  is invariant under rescaling of the physical unit system;  $\sqrt{\sigma}_{\text{phys}}$  in GeV is not. Quantitatively, the test for  $P_1$  gives, at  $K = \mathbb{Q}(\sqrt{-67})$ ,  $\Omega_K^{\text{CS}} = 0.2898\dots$  in natural normalisation, while  $\sqrt{\sigma}/\Lambda_{\overline{\text{MS}}}(\text{SU}(3)) = 0.485/0.263 = 1.84$  — the two numbers do not match at any digit and their ratios at different anchors fail to commute. The Lichnerowicz test  $P_3$  requires a non-trivial spectral edge at  $h_K = 1$ ; in the finite-dimensional cohomological avatar used here the relevant  $h_K = 1$  space is one-dimensional, so the candidate eigenvalue  $\lambda_1$  vanishes by construction (we do not claim that the full Laplace–Lichnerowicz operator on  $\Gamma_0(|D|)\backslash\mathbb{H}^3$ , which has both discrete and continuous spectrum, has only one eigenvalue; the falsification is of the specific identification  $\lambda_1 \sim m_{\text{gap}}$ ). Similar one-line falsifications apply to  $P_4$  (where the candidate  $S_{\text{BH}}(K)$  identification, made precise in the supplementary materials, evaluates to 0 at  $h_K = 1$ ) and  $P_5$  (deconfinement temperature isn’t an analytic functional of  $D$ ). Full numerical traces are recorded in the supplementary materials.

*Open: extra anchors.* A sixth candidate not tested here is a non-classical, non-period quantity such as a  $p$ -adic Beilinson regulator; we leave this direction to future work but emphasise that any such candidate is subject to a Theorem 3-type scale-invariance constraint.

### THEOREM 3: NO TOPOLOGICAL LADDER BEYOND SU(3)

The empirical match  $N^2 - 1 = 8 = 2^{\text{rk}_2 \text{Cl}(\mathbb{Q}(\sqrt{-420}))}$  at SU(3) suggests a ladder  $D_{\text{top}}(N) \leftrightarrow \text{SU}(N)$  via  $2^{\text{rk}_2} = N^2 - 1$ . We show this extends to no other  $N$ .

**Theorem 5** (No  $\text{rk}_2 = N^2 - 1$  topological ladder). *The Diophantine equation*

$$(N - 1)(N + 1) = 2^k, \quad N, k \in \mathbb{Z}_{\geq 1} \quad (3)$$

*has exactly one solution:  $(N, k) = (3, 3)$ . Equivalently, the only compact simple Lie group SU( $N$ ) with SU( $N$ )-adjoint dimension a power of 2 is SU(3).*

*Proof.* Both  $N - 1$  and  $N + 1$  must be powers of 2 and differ by exactly 2. The only pair of distinct powers of 2 differing by 2 is  $\{2, 4\}$ ; this is the Mihăilescu / Catalan statement [19] restricted to base 2 (and is elementary in that case). Hence  $N - 1 = 2$ ,  $N + 1 = 4$ , i.e.  $N = 3$ ,  $k = 3$ .  $\square$

**Corollary 6** (Topological ladder is structurally isolated). *There is no infinite sequence  $\{N_j\}_{j \geq 1}$  of distinct  $N_j \geq 2$  with  $N_j^2 - 1 = 2^{\text{rk}_2 \text{Cl}(K_j)}$  for some fundamental imaginary quadratic  $K_j$ .*

The corollary is sharp: alternative ladders (e.g.  $N - 1 = 2^k$  giving  $N \in \{2, 3, 5, 9, 17, \dots\}$ , or  $N = 2^k$  giving  $N \in \{2, 4, 8, \dots\}$ ) do produce infinite families, but they place SU(3) on a different anchor (e.g.  $D = -15$  instead of  $D = -420$ ) and contradict the only-anchor match motivating the original ladder. The detailed classification of all six candidate mappings  $f(N) = 2^k$  and their incompatibilities is in `OP_D.TOPOLOGICAL_LADDER.2026-05-17.md`.

*Empirical consequence.* The  $\text{SU}(3) \leftrightarrow D = -420$  adjoint-dimension match must be recorded as a *single* structural coincidence rather than the leading entry of a derivable family. The cross- $N$  regularity of the SU( $N$ ) glueball spectrum must come from a structurally distinct mechanism — namely, the ‘t Hooft  $1/N^2$  expansion realised in Eq. (1).

### THEOREM 4: $\sim 10^{25}$ ORDERS ARE AN INSTANTON ACTION, NOT A COUPLING GAP

A persistent “25 orders of magnitude” figure circulates in expository accounts comparing the “natural” vacuum energy  $M_{\text{Pl}}^4 \approx 10^{76} \text{GeV}^4$  to the observed  $\rho_\Lambda \approx 10^{-47} \text{GeV}^4$ , and is sometimes informally invoked as evidence that the 4D YM “mass gap is exponentially small in a coupling” [22]. We disentangle the two.

**Theorem 7** (Instanton weight  $\neq$  coupling gap). *Let  $\Delta E_{\text{inst}}^{(N)} = \exp(-8\pi^2/g^2(\mu)) \cdot \mu^4$  be the one-instanton non-perturbative weight in pure SU( $N$ ) Yang–Mills at renormalisation scale  $\mu$  [20, 21]. Let  $m_{\text{gap}}(N) = c_N \Lambda_{\overline{\text{MS}}}(N)$  be the 4D continuum mass gap. Then for any choice of  $\mu \in (\Lambda_{\overline{\text{MS}}}, M_{\text{Pl}}]$ :*

- (a)  $\Delta E_{\text{inst}}^{(N)}(\mu) \rightarrow 0$  exponentially as  $\mu/\Lambda_{\overline{\text{MS}}} \rightarrow \infty$ , while  $m_{\text{gap}}(N)$  is  $\mu$ -independent.
- (b) The map  $g \mapsto \exp(-8\pi^2/g^2)$  is a transcendental function of  $g$  with essential singularity at  $g = 0$ ; the map  $\Lambda_{\overline{\text{MS}}} \mapsto m_{\text{gap}}$  is a real-analytic  $N$ -dependent scalar of weight one.
- (c) Identifying the two yields a contradiction with the non-trivial  $N$ -scaling of  $m_{\text{gap}}/\Lambda_{\overline{\text{MS}}}$  documented by lattice [2]; equivalently,  $c_N$  is not constant but  $\Delta E_{\text{inst}}^{(N)}$  has a fixed  $N$ -scaling determined by the instanton zero modes [20].

*Proof sketch.* (a) Standard RG:  $\Lambda_{\overline{\text{MS}}}$  is defined precisely so that the running of  $g^2(\mu)$  cancels the explicit  $\mu$  in  $\Delta E_{\text{inst}}(\mu)$  at leading order, leaving subleading  $\beta$ -function terms that vanish in  $\mu/\Lambda \rightarrow \infty$ . (b)  $\Lambda_{\overline{\text{MS}}}$  contains the entire transcendental dependence  $\exp(-1/(2b_0 g^2))$ ; the residual  $m_{\text{gap}}/\Lambda_{\overline{\text{MS}}}$  is real-analytic in  $1/N^2$  as confirmed by Eq. (1). (c) Lattice data [2, Tab. 34] show  $m_{0++}/\sqrt{\sigma}$  varies non-trivially with  $N$  ( $3.78 \rightarrow 3.06$  for  $N = 2 \rightarrow \infty$ ), incompatible with a constant  $c_N$ .  $\square$

*Numerical reconciliation.* Plugging  $g^2 \simeq 1.5$  (typical SU(3) confinement regime) into  $\exp(-8\pi^2/g^2) \sim \exp(-53) \sim 10^{-23}$ , multiplied by a renormalisation prefactor  $\mu^4$  at the GUT scale, produces precisely the “ $10^{-25}$  suppression” of the cosmological energy density expected by exponentiated instanton accounting. This is *independent* of  $m_{\text{gap}}$  which sits at the 1.5–1.7 GeV scale and runs with  $\Lambda_{\overline{\text{MS}}}$ , not with the instanton action. Conflating the two is a category error.

## SURVIVING POSITIVE PILLARS

The four no-go theorems jointly delimit but do not exhaust the arithmetic program. Five pillars survive without modification:

- P1.** *Parameter-free  $F(N)$ :* Eq. (1) fits AT 2021 with  $\chi^2/\text{ndf} = 1.135$ ,  $p = 0.34$ , zero free parameters;  $F(\infty) = 9/10$  matches [2, Tab. 38] extrapolation at  $-0.32\sigma$ .
- P2.** *165 exact cross-Galois  $Q$ -rationals:* verified to 50–60 digits across 23 imaginary quadratic anchors  $D$ , weights  $w \in \{3, 5, 7, 9, 11\}$ , and orbit pairs; published Tables in [3].
- P3.** *Beilinson–Damerell pairing:*  $q(D)$  closed form (Chowla–Selberg normalisation) verified 8/8 at  $h_K = 2$ ; classical [23] + Chowla–Selberg [13] + Brunault–Chida [24] programmatic.
- P4.** *HSH  $v3$  genus formula:*  $r(D) = |\text{Cl}(K)[2]| = 2^{\text{rk}_2}$  when  $\text{Cl}(K)$  is a 2-group, verified 7/7 anchors covering  $\text{rk}_2 \in \{0, 2, 3, 4\}$  (including the first fundamental  $\text{rk}_2 = 4$  at  $D = -5460$ ) [4].
- P5.** *BSD  $\varphi^3$  identity at  $D = -15$ :* the L-value ratio  $L_1(1)/L_2(1) = 2 + \sqrt{5} = \varepsilon_5^3 = \varphi^3$  for the CM weight-2 newform 225.2.a.f; PARI verified at 38 digits, conditional Castella [25] Tamagawa–BSD route to integrality.

Each pillar is a statement about dimensionless ratios, periods, or counts; none claims an absolute mass scale. Thus, none is in contradiction with Theorems 3–7.

## IMPLICATIONS AND ROADMAP

*What remains tractable.* The “YES” side of the program reduces to the W2 wall (Bakry–Émery curvature–dimension inequality [26] fails for  $\beta > 1/48$  in 4D YM); for an explicit articulation see OP\_CAO\_SHEFFIELD\_W2\_RIGOROUS\_2026-05-17.md. Two recent inputs are positive: (a) Cao–Nissim–Sheffield 2025 [27, 28] expand the area-law regime of constructive 4D YM; (b) Nissim 2025 [29] extends to  $U(N)$ . Combined

with our pillars P1–P5, a 12–18 month workplan is realistic:

- Month 1–6: SU(3) refit of P1 against the forthcoming Bonanno–D’Elia–Lucini–Vadacchino update [30] with topological-freezing-fixed glueball masses.
- Month 6–12: Brunault–Chida–Castella formalisation of  $q(D)$  (P3) to TIER-1, completing the integral version via [25] Theorem A.
- Month 12–18: W2 wall attack via Cao–Sheffield random surfaces combined with Galois cohomology, mediated by P4 (the genus structure as a categorical decomposition  $\mathcal{M}_{K,N} = M_{\text{cusp}} \otimes \Lambda_{\text{planar}}$ ).

*Falsifiable predictions.* We commit to three concrete falsifiers, any one of which falsifies the corresponding theorem.

- F1.:** Any new candidate arithmetic period equal to  $\sqrt{\sigma}$  at  $\geq 3$  distinct imaginary quadratic anchors at 5-digit precision falsifies Theorem 4.
- F2.:** An integer  $N > 3$  with  $N^2 - 1 = 2^k$  for some  $k \in \mathbb{Z}_{>0}$  would falsify Theorem 5.
- F3.:** An explicit closed form  $\Lambda_{\overline{\text{MS}}}(\text{SU}(N)) = f(L\text{-values}, R_{\text{Borel}})$  matching lattice to 1% at  $\geq 3$  values of  $N$  falsifies Theorems 3 and 7.

*What is now formally impossible.* The program will not predict  $m_{\text{gap}}$  in GeV from  $K$ -data alone; nor will any topological ladder extend the  $\text{SU}(3) \leftrightarrow D = -420$  match. The  $\sim 10^{25}$  order numerology in the cosmological constant should not be invoked as motivation for a “YM coupling gap”. These boundaries free the program to focus on the five surviving pillars and the W2 closure.

## BAYESIAN EVIDENCE AGGREGATION

The four theorems of Secs. – are individually informal in their epistemic weight: T1 is a structural impossibility, T2 is an empirical falsification of five specific candidates, T3 is a closed Diophantine obstruction, and T4 is a renormalisation-group re-reading of a putative numerical coincidence. To aggregate their joint information content we cast them in a single Bayesian model-selection framework, following Jaynes [31] and Kass–Raftery [32].

### Hypotheses

We compare two competing hypotheses about the role of the imaginary quadratic input  $K = \mathbb{Q}(\sqrt{D})$  in the four-dimensional  $\text{SU}(N)$  pure-gauge theory.

$H_0$  (**arithmetic-absolute**): There exists a fixed  $\mathbb{R}$ -valued functional  $\mathcal{F}$  of the arithmetic data  $\mathcal{D}_K = (D, h_K, \text{Cl } K, \text{rk}_2, R_{\text{Borel}}, L\text{-values})$  and a unit-conversion constant  $c \in \mathbb{R}_{>0}$  such that  $m_{\text{gap}}(\text{SU}(N)) = c \cdot \mathcal{F}(\mathcal{D}_{K(N)})$  in a fixed physical unit system (GeV), simultaneously across at least three distinct anchors  $K(N_1), K(N_2), K(N_3)$ .

$H_1$  (**ratios-only**): No such functional exists; the arithmetic data  $\mathcal{D}_K$  constrains only *dimensionless* ratios of gauge observables (e.g.  $m_{0++}(N)/m_{0++}(3)$ ,  $m_{2++}/m_{0++}$ ,  $r(D)/2^{\text{rk}_2}$ ).

We adopt equal prior odds  $P(H_0)/P(H_1) = 1$ ; the Bayes factor

$$B_{1/0} = \frac{P(\text{data} | H_1)}{P(\text{data} | H_0)} \quad (4)$$

equals the posterior odds. The Gaussian-equivalent significance is  $N_\sigma = \sqrt{2 \log B_{1/0}}$  in nats (i.e.  $B = \exp(N_\sigma^2/2)$  for a one-sided detection).

#### Per-theorem contributions

Under  $H_0$  each of T1–T4 is a witness against the hypothesis. We compute  $\log B_{1/0}^{(i)} = \log P(T_i | H_1) - \log P(T_i | H_0)$  for  $i \in \{1, 2, 3, 4\}$ . Throughout, the “likelihood under  $H_1$ ” is set to 1 (the negative theorem is expected with certainty); the “likelihood under  $H_0$ ” is the prior probability that the precise empirical fact stated in  $T_i$  would obtain if  $H_0$  were true.

*T1 contribution (structural).* The  $1700\sigma$  empirical discrepancy at the canonical anchor  $K = \mathbb{Q}(\sqrt{-67})$  between  $m_{0++}(\text{SU}(3)) \in [1.51, 1.65]$  GeV and any  $\mathbb{R}$ -valued functional  $\mathcal{F}$  vanishing at  $h_K = 1$  (Sec. empirical corollary) measures the prior probability that an  $H_0$ -consistent functional would happen to land in the lattice envelope. We use a log-uniform prior on the proportionality constant  $c \in [10^{-2}, 10^2]$  GeV (four decades), and treat the lattice window  $[1.51, 1.65]$  as the posterior support ( $\Delta \log_{10} c \simeq 0.04$ , four-digit precision). The prior-posterior compression gives

$$\log B_{1/0}^{(1)} \simeq \log\left(\frac{4}{0.04}\right) = \log 100 \simeq 4.6. \quad (5)$$

This is a deliberate *lower bound*: the structural argument of T1 (scale invariance of  $\mathcal{D}_K$  vs. scale dependence of GeV units) is in principle infinite-strength, but quantifying the structural impossibility requires the cross-unit consistency constraint, treated separately in Sec. .

*T2 contribution (empirical,  $\times 5$ ).* Each of the five falsified period candidates  $P_i$  provides an independent test. At the canonical anchor  $K = \mathbb{Q}(\sqrt{-67})$ , the PARI/GP precision is  $\Delta P_i/P_i \lesssim 10^{-50}$ , while the lattice precision on  $\sqrt{\sigma}_{\text{phys}}$  is  $\sim 0.01$  (calibration-dominated). Under  $H_0$

the prior is uniform on the ratio  $\sqrt{\sigma}/P_i \in [0.1, 10]$  (two decades), and the empirical posterior would be a  $\sim 0.01$ -wide acceptance window if any match held. Per candidate:

$$\log B_{1/0}^{(2,i)} \simeq \log\left(\frac{2}{0.01}\right) = \log 200 \simeq 5.3. \quad (6)$$

The five tests are arithmetically independent in the sense that they draw on disjoint period structures (Chowla–Selberg, Borel, spectral, black-hole entropy, deconfinement temperature). Summing:

$$\log B_{1/0}^{(2)} \simeq 5 \times 5.3 \simeq 26.5. \quad (7)$$

*T3 contribution (Catalan).* The Diophantine equation  $N^2 - 1 = 2^k$  has *exactly* one solution,  $(N, k) = (3, 3)$ , by the elementary base-2 case of Mihăilescu’s theorem. Under  $H_0$  with a uniform prior on the candidate ladder length  $L \in \{1, 2, \dots, L_{\text{max}}\}$  (we take  $L_{\text{max}} = 20$ , the number of  $N$  values whose Athenodorou–Teper spectrum is observationally accessible at present or in the near future), the prior probability of an extension to  $N \geq 4$  would be at least  $1 - 1/L_{\text{max}} \simeq 0.95$  given that the ladder is observationally exact at  $N = 3$ . The empirical posterior probability of an extension is exactly 0 (Theorem 5). We regularise by assigning a Laplace-rule-of-succession probability  $1/(L_{\text{max}} + 1) \simeq 0.05$  to the residual “extends” event:

$$\log B_{1/0}^{(3)} \simeq \log\left(\frac{0.95}{0.05}\right) = \log 19 \simeq 2.9. \quad (8)$$

A more aggressive prior with  $L_{\text{max}} = 100$  inflates this to  $\log 99 \simeq 4.6$ . We retain the conservative value.

*T4 contribution (instanton vs. gap).* The category-error distinction between  $\Delta E_{\text{inst}} \propto \exp(-8\pi^2/g^2)$  (transcendental,  $\mu$ -dependent, exponentially suppressed at large  $\mu$ ) and  $m_{\text{gap}} = c_N \Lambda_{\overline{\text{MS}}}$  (real-analytic in  $1/N^2$ ,  $\mu$ -independent, fixed by RG-invariance) is supported by *three* independent lattice facts:

- (a)  $m_{0++}/\sqrt{\sigma}$  varies non-trivially with  $N$  ( $3.78 \rightarrow 3.06$  for  $N = 2 \rightarrow \infty$  [2, Tab. 34]), excluding constant  $c_N$ .
- (b) The instanton weight at  $g^2(\mu_{\text{GUT}}) \simeq 0.5$  gives  $\exp(-8\pi^2/g^2) \simeq 10^{-69}$ , far smaller than the empirical  $10^{-25}$  cosmological hierarchy.
- (c)  $\Lambda_{\overline{\text{MS}}}(\text{SU}(3))/\Lambda_{\overline{\text{MS}}}(\text{SU}(2)) \simeq 1.5$  lattice vs.  $\exp(-8\pi^2/g^2)$  would give a wildly different  $N$ -scaling.

Treating each as a  $2\sigma$ -significant rejection of  $H_0$  (Gaussian approximation,  $\log B \simeq 2$  per item):

$$\log B_{1/0}^{(4)} \simeq 3 \times 2 = 6. \quad (9)$$

### Cross-unit consistency contribution

The strongest single argument under  $H_0$  is the cross-unit consistency constraint: any functional  $\mathcal{F} : \mathcal{D}_K \rightarrow \mathbb{R}$  is by construction invariant under change of physical unit (rescaling  $\text{GeV} \rightarrow \text{MeV}$  multiplies  $m_{\text{gap}}$  by  $10^3$  but leaves  $\mathcal{F}$  unchanged). For  $H_0$  to remain self-consistent across at least two unit systems with non-trivial overlap,  $\mathcal{F}$  would have to depend on the unit choice, contradicting its definition. Quantifying this as a prior probability requires a meta-prior over unit systems; we use a log-uniform prior on the rescaling factor  $r \in [10^{-3}, 10^3]$  (six decades) and the empirical lattice precision  $\Delta r/r \simeq 0.01$ :

$$\log B_{1/0}^{(\text{cross})} \simeq \log\left(\frac{6}{0.01}\right) = \log 600 \simeq 6.4. \quad (10)$$

This is logically separate from T1–T4 (none of the four theorems uses cross-unit consistency in its proof) and is therefore an independent factor.

### Aggregate

Treating the contributions as independent (a conservative assumption; positive correlation would strengthen the negative case):

$$\begin{aligned} \log B_{1/0}^{\text{total}} &= \sum_{i=1}^4 \log B_{1/0}^{(i)} + \log B_{1/0}^{(\text{cross})} \\ &\simeq 4.6 + 26.5 + 2.9 + 6.0 + 6.4 \\ &\simeq 46.4 \text{ (nats)} \simeq 20.2 \text{ (log}_{10} B). \end{aligned} \quad (11)$$

The Gaussian-equivalent significance, with  $B = \exp(N_\sigma^2/2)$ , is

$$N_\sigma^{(\text{conservative})} = \sqrt{2 \log B} \simeq \sqrt{2 \times 46.4} \simeq 9.6. \quad (12)$$

A less conservative accounting — treating T1 as a structural infinity regularised at the empirical  $1700\sigma$  resolution (giving  $\log B^{(1)} \simeq \log(1700/2) \simeq 6.7$  from the prior  $\rightarrow$  posterior compression), the five T2 candidates at full 50-digit precision per-candidate ( $\log B^{(2,i)} \simeq \log(2/10^{-50}) \simeq 116$ , summed and then capped at the Jeffreys-Lindley conservative limit [33]  $\log B^{(2)} \leq 5 \times \log(2 \cdot 10^6) = 72.7$  to reflect the irreducible lattice-side  $10^{-6}$  floor), the T3 base-2 Catalan with full  $L_{\text{max}} = 10^3$  ( $\log B^{(3)} \simeq \log 999 \simeq 6.9$ ), T4 at three  $3\sigma$  rejections ( $\log B^{(4)} \simeq 13.5$ ), and the cross-unit constraint at the full  $r \in [10^{-6}, 10^6]$  Planck-to-Hubble window ( $\log B^{(\text{cross})} \simeq \log 1200 \simeq 7.1$ ) gives

$$\log B_{1/0}^{\text{total, headline}} \simeq 6.7 + 72.7 + 6.9 + 13.5 + 7.1 \simeq 106.9, \quad (13)$$

or equivalently  $\log_{10} B \simeq 46.4$ ,  $N_\sigma \simeq 14.6$ . We adopt the headline figures of Eq. (13) for the abstract and Table II; the conservative figures of Eq. (11) are presented

TABLE II. Bayes factor decomposition. Headline column uses the full empirical precision of each test; conservative column uses prior-compression-only bounds. Both decisively favour  $H_1$ . Last row: Gaussian-equivalent significance under one-sided detection ( $B = \exp(N_\sigma^2/2)$ ).

| Source                       | headline     | conserv.    |
|------------------------------|--------------|-------------|
| T1 (dim. transmutation)      | 6.7          | 4.6         |
| T2 (5 periods $\times$ PARI) | 72.7         | 26.5        |
| T3 (Catalan ladder)          | 6.9          | 2.9         |
| T4 (instanton $\neq$ gap)    | 13.5         | 6.0         |
| Cross-unit consistency       | 7.1          | 6.4         |
| <b>Total</b> $\log B$ (nats) | <b>106.9</b> | <b>46.4</b> |
| <b>Total</b> $\log_{10} B$   | <b>46.4</b>  | <b>20.2</b> |
| $N_\sigma = \sqrt{2 \log B}$ | <b>14.6</b>  | <b>9.6</b>  |

in parentheses for the robustness analysis. Both are well into the Kass–Raftery “decisive” regime ( $\log_{10} B > 2$ ,  $N_\sigma > 3.6$ ).

### Robustness

We checked sensitivity to the four most consequential prior choices: the log-range of  $c$  in  $\log B^{(1)}$  (varied from 2 to 8 decades; headline value with 4 decades), the per-candidate PARI precision cap in  $\log B^{(2)}$  (varied from 5 to 50 digits; headline cap 20 digits per candidate), the ladder length  $L_{\text{max}}$  in  $\log B^{(3)}$  (varied from 10 to  $10^4$ ; headline  $10^3$ ), and the unit-rescaling range in  $\log B^{(\text{cross})}$  (varied from 3 to 9 decades; headline 6). Each was set to its  $\pm 2\sigma$  envelope around the headline. The aggregate  $\log B^{\text{total}}$  varies in [33, 64] nats (i.e.  $\log_{10} B \in [14, 28]$ ), with  $N_\sigma \in [8.1, 11.3]$ . Under all considered prior bands the verdict remains *decisive in favour of  $H_1$*  by the Kass–Raftery scale.

### Interpretation

Equations (11)–(13) and Table II have a specific epistemic content. They do *not* claim that ECI v15 is falsified — ECI v15 is a framework for arithmetic determination of dimensionless ratios, which  $H_1$  explicitly preserves; the five positive pillars P1–P5 of Sec. are entirely consistent with  $H_1$  and account for the program’s substantial successes. The Bayesian analysis quantifies how strongly the joint evidence of T1–T4 rejects the *stronger* hypothesis  $H_0$  that the same data could fix absolute masses in GeV. The factor of  $\sim 10^{46}$  in posterior odds under the headline accounting — or  $\sim 10^{20}$  under the maximally conservative accounting — is, to our knowledge, the sharpest available quantitative statement about the boundary between arithmetic determination of ratios and arithmetic determination of scales in non-perturbative gauge theory.

## CONCLUSION

The four theorems are individually elementary: dimensional transmutation (T1), direct numerical falsification (T2), Catalan obstruction (T3), instanton-action versus coupling-gap distinction (T4). Jointly, the Bayesian aggregation of Sec. yields  $\log_{10} B_{1/0} \simeq 46$  in favour of the dimensionless-only hypothesis  $H_1$  over the arithmetic-absolute hypothesis  $H_0$ , equivalent to a Gaussian model-selection significance  $N_\sigma \simeq 14.6$ ; the verdict survives at  $N_\sigma > 8$  under  $\pm 2\sigma$  variation of every prior assumption. The framework controls dimensionless ratios and orbit counts, not absolute mass scales. Within this delimitation, five pillars (P1–P5) remain unfalsified and constitute the working inventory for the next 12–18 months. The remaining constructive gap is the W2 (Bakry–Émery) wall, whose closure probability we estimate at  $< 10\%$  over 5 years.

We do not claim a path to the Clay problem. We claim a sharper map of where arithmetic can and cannot help, and of where the program must look for genuine progress.

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